D.3. Joint decomposition and spectral localization

Because the Q_k commute, we have the joint decomposition $X = \bigoplus_{\varepsilon\in\{0,1\}^K} X_\varepsilon$, $X_\varepsilon := \left(\prod_{k=1}^K Q_k^{\varepsilon_k}(I-Q_k)^{1-\varepsilon_k}\right) X$, into closed, A-invariant subspaces. Write $\varepsilon = (\varepsilon_1, \dots, \varepsilon_K)$. On X_ε, $\sigma(\phi_k(A)|_{X_\varepsilon}) \subset \begin{cases} \{1\}, & \varepsilon_k = 1, \\ \{\zeta : |\zeta - 1| \geq r_k\}, & \varepsilon_k = 0. \end{cases}$ By spectral mapping for holomorphic calculus (used on p. 7), for the restricted operator $A|_{X_\varepsilon}$ one has $\sigma(\phi_k(A)|_{X_\varepsilon}) = \phi_k(\sigma(A|_{X_\varepsilon}))$, hence

$$\sigma(A|_{X_\varepsilon}) \subset \{\lambda \in \sigma(A) : \phi_k(\lambda) = 1 \text{ if } \varepsilon_k = 1, \phi_k(\lambda) \neq 1 \text{ if } \varepsilon_k = 0\}. \tag{2}$$

(This is precisely the localization corresponding to the preimages $\phi_k^{-1}(\{1\})$ selected by the Riesz projections.)

D.4. Invertibility of $\Psi(A) - I$ off the joint 1-eigenspace

Lemma .18 (Separation $\implies$ no unit spectrum off the joint block)**.** *Assume (D2). Then for every $\varepsilon \neq (1, \dots, 1)$, $1 \notin \sigma(\Psi(A)|_{X_\varepsilon})$.*

Proof. Fix $\varepsilon \neq (1, \dots, 1)$. By spectral mapping (again for restrictions), $\sigma(\Psi(A)|_{X_\varepsilon}) = \Psi(\sigma(A|_{X_\varepsilon}))$. If $1 \in \sigma(\Psi(A)|_{X_\varepsilon})$, then there exists $\lambda \in \sigma(A|_{X_\varepsilon})$ with $\Psi(\lambda) = 1$. By (D1)–(D3) and the construction of X_ε, (2) forces $\phi_k(\lambda) = 1$ whenever $\varepsilon_k = 1$, and $\phi_k(\lambda) \neq 1$ whenever $\varepsilon_k = 0$. This contradicts (D2) unless $\varepsilon = (1, \dots, 1)$. Thus $1 \notin \sigma(\Psi(A)|_{X_\varepsilon})$. □

As a consequence, $\Psi(A) - I$ is boundedly invertible on $\bigoplus_{\varepsilon\neq(1,\dots,1)} X_\varepsilon$.

D.5. Identification of the fixed space and of the Riesz projection of $\Psi(A)$

Theorem .19 (Corrected Lemma 3.3; Banach H^∞-calculus version)**.** *Under (D1)–(D3) and (1), one has* $\ker(I - \Psi(A)) = \bigcap_{k=1}^K \ker(I - \phi_k(A))$. *Moreover, if P_Ψ denotes the Riesz projection of $\Psi(A)$ at 1 (which exists when 1 is isolated in $\sigma(\Psi(A))$, e.g. under Theorem 3.1(b)), then $P_\Psi = \prod_{k=1}^K Q_k$, hence* $\operatorname{Ran} P_\Psi = \bigcap_{k=1}^K \operatorname{Ran} Q_k = \bigcap_{k=1}^K \ker(I - \phi_k(A))$.

Proof. ($\subseteq$): If $x \in \ker(I - \phi_k(A))$ for all k, then $\phi_k(A)x = x$ and, by commutativity of the layers (each is a function of A, Section 3 p. 6), $\Psi(A)x = x$. Thus $\bigcap_k \ker(I - \phi_k(A)) \subseteq \ker(I - \Psi(A))$.

($\supseteq$): Decompose $x = \sum_\varepsilon x_\varepsilon$ with $x_\varepsilon \in X_\varepsilon$. If $x \in \ker(I - \Psi(A))$, then $0 = (\Psi(A) - I)x = \sum_\varepsilon (\Psi(A) - I)x_\varepsilon$. By Lemma .18, $\Psi(A) - I$ is invertible on X_ε for $\varepsilon \neq (1, \dots, 1)$; hence $x_\varepsilon = 0$ for all such ε. Therefore $x = x_{(1,\dots,1)} \in X_{(1,\dots,1)} = \bigcap_k \operatorname{Ran} Q_k$. By Lemma .17, $\operatorname{Ran} Q_k = \ker(I - \phi_k(A))$, so $x \in \bigcap_k \ker(I - \phi_k(A))$. This proves the equality of fixed spaces.

For the projection identity, note that: (i) $\Psi(A) - I$ is invertible on $\bigoplus_{\varepsilon\neq(1,\dots,1)} X_\varepsilon$ (Lemma .18); (ii) $\Psi(A) = I$ on $X_{(1,\dots,1)}$ because $\phi_k(A) = I$ there. Hence the Riesz projection P_Ψ at 1 is precisely the projection onto $X_{(1,\dots,1)}$ along $\bigoplus_{\varepsilon\neq(1,\dots,1)} X_\varepsilon$, i.e. $P_\Psi = \prod_{k=1}^K Q_k$, since the product of commuting idempotents projects onto the intersection of their ranges. The range equalities follow from Lemma .17. □

D.6. Consequences for Theorem 3.1 and consistency with Section 3

Range identification (Remark after Lemma 3.3). Under (D1)–(D3) and (1), the mean-ergodic projection P from Theorem 3.1(a) equals the Riesz projection P_Ψ at 1 (when 1 is isolated as in Theorem 3.1(b)), and by Theorem .19 its range is exactly $\bigcap_k \ker(I - \phi_k(A))$. This replaces the spectral-measure argument sketched after Lemma 3.3 (pp. 7–8).

Boundary case of (1). If each ϕ_k has the peripheral fixed-point property, Appendix C, Proposition .13 proves (1) for boundary spectral values; this supplies a widely applicable sufficient condition for (D2) at the spectral boundary.

Power-boundedness used implicitly. The step $\operatorname{Ran} Q_k = \ker(I - \phi_k(A))$ invoked in Section 3 now follows from Lemma .17 together with (D3), aligning with the mean-ergodic discussion in Theorem 3.1(a) (p. 6–7) and the calculus bound in Lemma .12 (Appendix C.2, p. 13).

D.7. Sectorial (continuous-time) variant

The same reasoning applies when $-A$ generates a bounded analytic semigroup and each ϕ_k is holomorphic on a sector containing $\sigma(A)$, with $\phi_k(1) = 1$. Isolation and separation are imposed at the value 1 in $\sigma(\phi_k(A))$, exactly as in (D1)–(D2). The construction of Q_k, the joint decomposition, Lemma .17 (power-boundedness of $\phi_k(A)$ from the H^∞ calculus) and Theorem .19 carry over verbatim. This is consistent with Section 3 (continuous-time branch of Theorem 3.1, p. 6–7).

Summary. Under the explicit isolation (D1), separation (1) (D2), and power-boundedness (D3), the fixed-space identity $\ker(I - \Psi(A)) = \bigcap_{k=1}^{K} \ker(I - \phi_k(A))$ holds in the Banach H^∞-calculus setting, and the Riesz projection of $\Psi(A)$ at 1 equals $\prod_{k=1}^{K} Q_k$. This provides a fully Banach-appropriate replacement for the spectral-measure step sketched after Lemma 3.3 and aligns with the hypotheses and uses of Theorem 3.1 (pp. 6–8) and Appendix C (pp. 12–13).

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